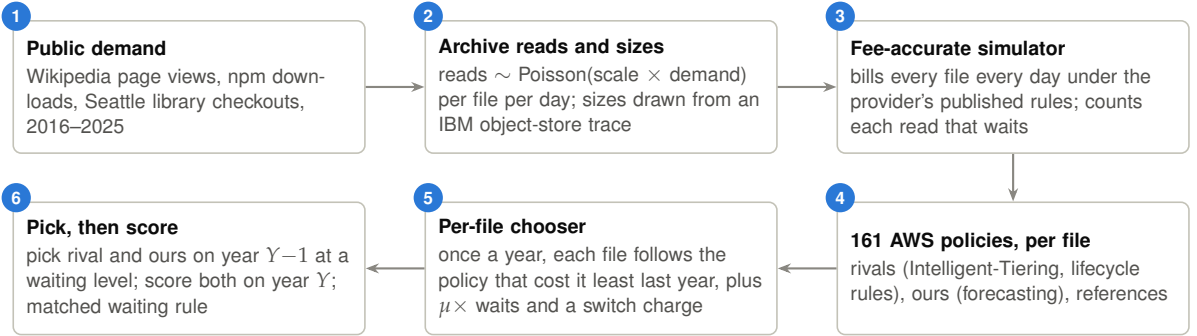
1
Public demand
Wikipedia page views, npm downloads, Seattle library checkouts, 2016–2025
2
Archive reads and sizes
reads ∼ Poisson(scale × demand) per file per day; sizes drawn from an IBM object-store trace
3
Fee-accurate simulator
bills every file every day under the provider's published rules; counts each read that waits
4
161 AWS policies, per file
rivals (Intelligent-Tiering, lifecycle rules), ours (forecasting), references
5
Per-file chooser
once a year, each file follows the policy that cost it least last year, plus $\mu\times$ waits and a switch charge
6
Pick, then score
pick rival and ours on year $Y-1$ at a waiting level; score both on year $Y$; matched waiting rule